Fall 2025 CPTS530 Homework 4
MATH 448-548
Numerical Analysis

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Problem 1

Problem Statement (Textbook 4.1.7):

Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $(B - I)$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

Solution:

To understand the pattern, let's first compute A^2 and A^3 directly.

Computing A^2:

$$\begin{aligned} A^2 &= \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

Computing A^3:

$$\begin{aligned} A^3 = A^2 \cdot A &= \begin{bmatrix} B^2 & BC + C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^3 & B^2C + BC + C \\ 0 & I \end{bmatrix} \end{aligned}$$

Observing the pattern in the upper-right block:

$$\begin{aligned} k = 1 : &\quad C \\ k = 2 : &\quad BC + C \\ k = 3 : &\quad B^2C + BC + C \end{aligned}$$

We can factor these as:

$$\begin{aligned} k = 1 : &\quad (I)C \\ k = 2 : &\quad (B + I)C \\ k = 3 : &\quad (B^2 + B + I)C \end{aligned}$$

This suggests the general formula: the upper-right block is $(I + B + B^2 + \cdots + B^{k-1})C$, which equals $(B^k - I)(B - I)^{-1}C$ (see Appendix A).

Now we prove this result by induction on k.

Base Case ($k = 1$):
For $k = 1$, we need to verify that

$$A^1 = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} = \begin{bmatrix} B^1 & (B^1 - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

This simplifies to showing that $C = (B - I)(B - I)^{-1}C$, which is trivially true since $(B - I)(B - I)^{-1} = I$.
Inductive Step:
Assume the formula holds for $k = n$:

$$A^n = \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

We need to prove it holds for $k = n + 1$. Computing $A^{n+1} = A^n \cdot A$:

$$\begin{aligned} A^{n+1} &= \begin{bmatrix} B^n & (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} B^n \cdot B + (B^n - I)(B - I)^{-1}C \cdot 0 & B^n \cdot C + (B^n - I)(B - I)^{-1}C \cdot I \\ 0 \cdot B + I \cdot 0 & 0 \cdot C + I \cdot I \end{bmatrix} \\ &= \begin{bmatrix} B^{n+1} & B^nC + (B^n - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix} \end{aligned}$$

Now we need to show that the upper-right block equals $(B^{n+1} - I)(B - I)^{-1}C$.
Key Identity:
We claim that $(B^{n+1} - I)(B - I)^{-1}C = B^nC + (B^n - I)(B - I)^{-1}C$.
This identity is equivalent to showing that:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

The proof of this geometric series identity for matrices is provided in Appendix A.
Thus, we have shown that:

$$A^{n+1} = \begin{bmatrix} B^{n+1} & (B^{n+1} - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

By the principle of mathematical induction, the formula holds for all $k \geq 1$.

Problem 2

Problem Statement (Textbook 4.1.12):

Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

Solution: (Approach followed as per lecture notes [1])

Let $\tilde{A} = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$ where A is $n \times n$ invertible, and $u, v \in \mathbb{R}^n$.

Part 1: Necessary and Sufficient Condition for Invertibility

For $\tilde{A}$ to be invertible, we need $\tilde{A}x = 0 \Rightarrow x = 0$.

Write $x = \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix}$ where $\bar{x} \in \mathbb{R}^n$ and $x_n \in \mathbb{R}$. Then:

$$\tilde{A}x = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix} \begin{bmatrix} \bar{x} \\ x_n \end{bmatrix} = \begin{bmatrix} A\bar{x} + x_n u \\ v^T \bar{x} \end{bmatrix} = 0$$

This gives us:

$$A\bar{x} + x_n u = 0 \tag{1}$$
$$v^T \bar{x} = 0 \tag{2}$$

From (1): $\bar{x} = -x_n A^{-1} u$ (since A is invertible).

Substituting into (2): $v^T(-x_n A^{-1} u) = 0$, which gives $-x_n(v^T A^{-1} u) = 0$.

If $s := v^T A^{-1} u \neq 0$, then $x_n = 0$, and from (1), $A\bar{x} = 0 \Rightarrow \bar{x} = 0$ (since A is invertible).

Therefore, $\tilde{A}$ is invertible if and only if:

$$\boxed{s = v^T A^{-1} u \neq 0}$$

Part 2: Formula for the Inverse

Seek $\tilde{A}^{-1} = \begin{bmatrix} B & \hat{u} \\ \hat{v}^T & k \end{bmatrix}$ where $B \in \mathbb{R}^{n \times n}$, $\hat{u}, \hat{v} \in \mathbb{R}^n$, $k \in \mathbb{R}$.

From $\tilde{A}\tilde{A}^{-1} = I_{n+1}$, block multiplication gives:

$$AB + u\hat{v}^T = I_n \tag{3}$$
$$A\hat{u} + ku = 0 \tag{4}$$
$$v^T B = 0^T \tag{5}$$
$$v^T \hat{u} = 1 \tag{6}$$

From (4): $\hat{u} = -kA^{-1}u$. Substituting into (6):

$$v^T(-kA^{-1}u) = 1 \Rightarrow -ks = 1 \Rightarrow k = -\frac{1}{s}$$

Therefore: $\hat{u} = -kA^{-1}u = \frac{1}{s}A^{-1}u$.
From (3): $B = A^{-1}(I_n - u\hat{v}^T)$. Substituting into (5):

$$v^T A^{-1}(I_n - u\hat{v}^T) = 0^T$$

$$v^T A^{-1} - (v^T A^{-1} u)\hat{v}^T = 0^T$$

$$v^T A^{-1} - s\hat{v}^T = 0^T$$

$$\hat{v}^T = \frac{1}{s} v^T A^{-1}$$

Therefore: $\hat{v} = \frac{1}{s}(A^{-1})^T v = \frac{1}{s}(A^T)^{-1}v$.
Finally, from $B = A^{-1}(I_n - u\hat{v}^T)$:

$$B = A^{-1}\left(I_n - u \cdot \frac{1}{s} v^T A^{-1}\right) = A^{-1} - \frac{1}{s}A^{-1}uv^T A^{-1}$$

Final Answer:
When $s = v^T A^{-1} u \neq 0$:

$$\boxed{\tilde{A}^{-1} = \begin{bmatrix} A^{-1} - \frac{1}{s}A^{-1}uv^T A^{-1} & \frac{1}{s}A^{-1}u \\ \frac{1}{s}v^T A^{-1} & -\frac{1}{s} \end{bmatrix}}$$

where $s = v^T A^{-1} u$.

Problem 3

Problem Statement (Textbook 4.1.17):

A square matrix A is said to be **skew-symmetric** if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x.

Solution:

Let A be a skew-symmetric matrix, so $A^T = -A$. Consider the scalar quantity $x^T Ax$.

Taking the transpose of this scalar (which equals itself):

$$\begin{aligned}(x^T Ax)^T &= x^T A^T (x^T)^T \\ &= x^T A^T x \\ &= x^T(-A)x \\ &= -x^T Ax\end{aligned}$$

Since $x^T Ax$ is a scalar and equals its own negative, we have $x^T Ax = -x^T Ax$, which implies $2x^T Ax = 0$.

Therefore, $x^T Ax = 0$ for all x. $\square$

Problem 4

Problem Statement (Textbook 4.1.18):

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

Solution:

Part 1: Diagonal elements are zero.

Consider a general skew-symmetric matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}$$

Since $A^T = -A$, comparing diagonal elements:

$$a_{11} = -a_{11}, \quad a_{22} = -a_{22}, \quad \ldots, \quad a_{nn} = -a_{nn}$$

For any diagonal element a_{ii}, we have $a_{ii} = -a_{ii}$, which implies:

$$a_{ii} = (A^T)_{ii} = (-A)_{ii} = -a_{ii}$$

This implies $2a_{ii} = 0$, so:

$$\boxed{a_{ii} = 0 \text{ for all } i}$$

Part 2: Determinant is zero for odd order.

Examples: Consider

$$A_3 = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{bmatrix} \text{ (odd order 3)}, \quad A_2 = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix} \text{ (even order 2)}$$

We have $\det(A_3) = 0$ while $\det(A_2) = 25 \neq 0$. This illustrates that the determinant is zero for odd order but not necessarily for even order.

General Proof:

For a skew-symmetric matrix A of order n, we have $A^T = -A$. Taking determinants:

$$\det(A^T) = \det(-A) = (-1)^n \det(A)$$

where we used $\det(\lambda A) = \lambda^n \det(A)$ with $\lambda = -1$ (see Appendix B.2).
Since $\det(A^T) = \det(A)$ (see Appendix B.1), we get:

$$\det(A) = (-1)^n \det(A)$$

If n is odd, then $(-1)^n = -1$, giving $\det(A) = -\det(A)$, which implies:

$$\boxed{\det(A) = 0 \text{ for odd order } n}$$

$\square$

Problem 5

Problem Statement (Textbook 4.1.19):

(Continuation) Let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x, $x^T Ax = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

Solution:
Let A be any $n \times n$ matrix, and define:

$$A_0 = \frac{1}{2}(A + A^T) \quad \text{and} \quad A_1 = \frac{1}{2}(A - A^T)$$

Part 1: A_0 is symmetric

$$A_0^T = \left[\frac{1}{2}(A + A^T)\right]^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = A_0$$

$$\boxed{A_0^T = A_0, \text{ so } A_0 \text{ is symmetric}}$$

Part 2: A_1 is skew-symmetric

$$A_1^T = \left[\frac{1}{2}(A - A^T)\right]^T = \frac{1}{2}(A^T - (A^T)^T) = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T) = -A_1$$

$$\boxed{A_1^T = -A_1, \text{ so } A_1 \text{ is skew-symmetric}}$$

Part 3: $A = A_0 + A_1$

$$A_0 + A_1 = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = \frac{1}{2}(A + A^T + A - A^T) = \frac{1}{2}(2A) = A$$

$$\boxed{A = A_0 + A_1}$$

Part 4: $x^T Ax = x^T A_0 x$ for all x
Since $A = A_0 + A_1$:

$$x^T Ax = x^T(A_0 + A_1)x = x^T A_0 x + x^T A_1 x$$

We need to show that $x^T A_1 x = 0$ for all x.
Note that $x^T A_1 x$ is a scalar (a 1×1 matrix), so it equals its own transpose:

$$(x^T A_1 x)^T = x^T A_1^T (x^T)^T = x^T A_1^T x$$

Since A_1 is skew-symmetric ($A_1^T = -A_1$):

$$(x^T A_1 x)^T = x^T(-A_1)x = -x^T A_1 x$$

Thus, $x^T A_1 x = -(x^T A_1 x)$, which implies $2(x^T A_1 x) = 0$, so $x^T A_1 x = 0$.
Therefore:

$$x^T A x = x^T A_0 x + 0 = x^T A_0 x$$

$$\boxed{x^T A x = x^T A_0 x \text{ for all } x}$$

Conclusion: This shows that any square matrix can be decomposed into symmetric and skew-symmetric parts, and the quadratic form $x^T A x$ depends only on the symmetric part A_0. This explains why we can restrict our attention to symmetric matrices when studying quadratic forms.

Problem 6

Part (a)

Problem Statement (Textbook 4.2.1a):

Prove these facts, needed in the proof of Theorem 2.

Problem 6(a) If U is upper triangular and invertible, then U^{-1} is upper triangular.

Solution:

Let U be upper triangular and invertible. To find U^{-1}, we solve $UX = I$ column by column. For the j-th column $\mathbf{x}_j$ of U^{-1}, we solve:

$$U\mathbf{x}_j = \mathbf{e}_j$$

where $\mathbf{e}_j$ is the j-th standard basis vector.

Since U is upper triangular with nonzero diagonal entries (by invertibility), this system can be solved by back substitution starting from row n. The key observation is:

- Row n: $u_{nn}x_{jn} = \delta_{jn}$ gives $x_{jn} = 0$ for all $j < n$
- Row $n-1$: $u_{n-1,n-1}x_{j,n-1} + u_{n-1,n}x_{jn} = \delta_{j,n-1}$ gives $x_{j,n-1} = 0$ for all $j < n-1$
- Continuing upward, row i depends only on $x_{ji}, x_{j,i+1}, \ldots, x_{jn}$

By induction, for $i > j$, we have $x_{ji} = 0$. Therefore, each column of U^{-1} has zeros below the diagonal, proving U^{-1} is upper triangular. $\square$

Example: Consider $U = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. To find U^{-1}, solve $U\mathbf{x}_j = \mathbf{e}_j$ for each column:

Column 1: $U\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{aligned} \text{Row 3:} \quad & 1 \cdot x_{31} = 0 \implies x_{31} = 0 \\ \text{Row 2:} \quad & 1 \cdot x_{21} + 2(0) = 0 \implies x_{21} = 0 \\ \text{Row 1:} \quad & 2 \cdot x_{11} + 1(0) + 3(0) = 1 \implies x_{11} = 1/2 \end{aligned}$$

Thus $\mathbf{x}_1 = [1/2, 0, 0]^T$ (zeros below diagonal).

Column 2: $U\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\begin{aligned} \text{Row 3:} \quad & x_{32} = 0 \\ \text{Row 2:} \quad & x_{22} + 2(0) = 1 \implies x_{22} = 1 \\ \text{Row 1:} \quad & 2x_{12} + 1(1) + 3(0) = 0 \implies x_{12} = -1/2 \end{aligned}$$

Thus $\mathbf{x}_2 = [-1/2, 1, 0]^T$ (zero in position $(3, 2)$).

Column 3: $U\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\begin{aligned} \text{Row 3:} \quad & x_{33} = 1 \\ \text{Row 2:} \quad & x_{23} + 2(1) = 0 \implies x_{23} = -2 \\ \text{Row 1:} \quad & 2x_{13} + 1(-2) + 3(1) = 0 \implies x_{13} = -1/2 \end{aligned}$$

Thus $\mathbf{x}_3 = [-1/2, -2, 1]^T$ (no zeros required below diagonal).

Therefore: $U^{-1} = \begin{bmatrix} 1/2 & -1/2 & -1/2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$ is upper triangular.

Part (b)

Problem Statement (Textbook 4.2.1b):
Prove these facts, needed in the proof of Theorem 2.
Problem 6(b) The inverse of a unit lower triangular matrix is unit lower triangular.

Solution:
Let L be unit lower triangular (lower triangular with ones on the diagonal). To find L^{-1}, we solve $LX = I$ column by column. For the j-th column $\mathbf{x}_j$ of L^{-1}, we solve:

$$L\mathbf{x}_j = \mathbf{e}_j$$

Since L is unit lower triangular, this system can be solved by forward substitution starting from row 1. The key observation is:

- Row 1: $x_{j1} = \delta_{j1}$ (so $x_{j1} = 1$ if $j = 1$, else $x_{j1} = 0$)
- Row 2: $l_{21}x_{j1} + x_{j2} = \delta_{j2}$ gives $x_{j2} = 0$ for all $j > 2$
- Continuing downward, row i depends only on $x_{j1}, x_{j2}, \ldots, x_{ji}$

By induction, for $i < j$, we have $x_{ji} = 0$ (zeros above diagonal). Moreover, since row j gives $l_{j1}x_{j1} + \cdots + l_{j,j-1}x_{j,j-1} + x_{jj} = 1$, we get $x_{jj} = 1$ (ones on diagonal). Therefore, L^{-1} is unit lower triangular. □

Example: Consider $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}$. To find L^{-1}, solve $L\mathbf{x}_j = \mathbf{e}_j$ for each column:

Column 1: $L\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{aligned} \text{Row 1:} \quad & 1 \cdot x_{11} = 1 \implies x_{11} = 1 \\ \text{Row 2:} \quad & 2(1) + 1 \cdot x_{21} = 0 \implies x_{21} = -2 \\ \text{Row 3:} \quad & 1(1) + 3(-2) + 1 \cdot x_{31} = 0 \implies x_{31} = 5 \end{aligned}$$

Thus $\mathbf{x}_1 = [1, -2, 5]^T$ (ones on diagonal, zeros above).

Column 2: $L\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\begin{aligned} \text{Row 1:} \quad & x_{12} = 0 \\ \text{Row 2:} \quad & 2(0) + x_{22} = 1 \implies x_{22} = 1 \\ \text{Row 3:} \quad & 1(0) + 3(1) + x_{32} = 0 \implies x_{32} = -3 \end{aligned}$$

Thus $\mathbf{x}_2 = [0, 1, -3]^T$ (zero in position $(1, 2)$, one on diagonal).

Column 3: $L\mathbf{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\begin{aligned} \text{Row 1:} \quad & x_{13} = 0 \\ \text{Row 2:} \quad & 2(0) + x_{23} = 0 \implies x_{23} = 0 \\ \text{Row 3:} \quad & 1(0) + 3(0) + x_{33} = 1 \implies x_{33} = 1 \end{aligned}$$

Thus $\mathbf{x}_3 = [0, 0, 1]^T$ (zeros above diagonal, one on diagonal).

Therefore: $L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -3 & 1 \end{bmatrix}$ is unit lower triangular.

Part (c)

Problem Statement (Textbook 4.2.1c):

Prove these facts, needed in the proof of Theorem 2.

Problem 6(c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

Solution:

Upper triangular case: Let U_1 and U_2 be upper triangular matrices. Consider entry (i, j) of their product U_1U_2:

$$(U_1U_2)_{ij} = \sum_{k=1}^{n}(U_1)_{ik}(U_2)_{kj}$$

For $i > j$, we need to show $(U_1U_2)_{ij} = 0$. In the sum above:

- If $k < i$: $(U_1)_{ik} = 0$ (since U_1 is upper triangular)
- If $k > j$: $(U_2)_{kj} = 0$ (since U_2 is upper triangular)
- If $i \leq k \leq j$: This range is empty when $i > j$

Therefore, every term in the sum is zero when $i > j$, so $(U_1U_2)_{ij} = 0$ for $i > j$, proving U_1U_2 is upper triangular.

Lower triangular case: The proof is analogous. For $i < j$, either $k > i$ (making $(L_1)_{ik} = 0$) or $k < j$ (making $(L_2)_{kj} = 0$), so $(L_1L_2)_{ij} = 0$. □

Example (Upper Triangular): Let

$$U_1 = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \quad U_2 = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Computing U_1U_2 element-by-element:
$(U_1U_2)_{11} = 2(1) + 1(0) + 3(0) = 2$
$(U_1U_2)_{12} = 2(2) + 1(2) + 3(0) = 6$
$(U_1U_2)_{13} = 2(1) + 1(3) + 3(1) = 8$
$(U_1U_2)_{21} = 0(1) + 1(0) + 2(0) = 0$ (zero below diagonal)
$(U_1U_2)_{22} = 0(2) + 1(2) + 2(0) = 2$
$(U_1U_2)_{23} = 0(1) + 1(3) + 2(1) = 5$
$(U_1U_2)_{31} = 0(1) + 0(0) + 1(0) = 0$ (zero below diagonal)
$(U_1U_2)_{32} = 0(2) + 0(2) + 1(0) = 0$ (zero below diagonal)
$(U_1U_2)_{33} = 0(1) + 0(3) + 1(1) = 1$

Therefore: $U_1U_2 = \begin{bmatrix} 2 & 6 & 8 \\ 0 & 2 & 5 \\ 0 & 0 & 1 \end{bmatrix}$ is upper triangular.

Problem 7

Problem Statement (Textbook 4.2.5):

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

Solution:

A Matrix Geometric Series

Lemma

For any square matrix B such that $(B - I)$ is nonsingular, and for any $k \geq 1$:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

Proof:
Define the partial sum sequence:

$$\begin{aligned}
S_0 &= 0 \\
S_1 &= I \\
S_2 &= I + B \\
S_3 &= I + B + B^2 \\
&\vdots \\
S_k &= I + B + B^2 + \cdots + B^{k-1}
\end{aligned}$$

We will show that $(B - I)S_k = B^k - I$ for all $k \geq 1$.
Multiplying S_k by $(B - I)$:

$$\begin{aligned}
(B - I)S_k &= (B - I)(I + B + B^2 + \cdots + B^{k-1}) \\
&= B(I + B + B^2 + \cdots + B^{k-1}) - (I + B + B^2 + \cdots + B^{k-1}) \\
&= (B + B^2 + B^3 + \cdots + B^k) - (I + B + B^2 + \cdots + B^{k-1}) \\
&= B^k - I
\end{aligned}$$

Since $(B - I)$ is nonsingular by assumption, we can multiply both sides by $(B - I)^{-1}$ on the left:

$$S_k = (B^k - I)(B - I)^{-1}$$

Therefore:

$$(B^k - I)(B - I)^{-1} = I + B + B^2 + \cdots + B^{k-1}$$

This completes the proof. $\square$

B Determinant Properties

B.1 Determinant of Transpose

Theorem

For any square matrix A, $\det(A^T) = \det(A)$.

Examples:
For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Direct computation:

$$\det(A) = ad - bc, \quad \det(A^T) = ad - cb = ad - bc$$

For a 3×3 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Expanding $\det(A)$ along the first row:

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expanding $\det(A^T)$ along the first column:

$$\det(A^T) = a_{11} \begin{vmatrix} a_{22} & a_{32} \\ a_{23} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{31} \\ a_{22} & a_{32} \end{vmatrix}$$

Each 2×2 minor in $\det(A^T)$ is the transpose of the corresponding minor in $\det(A)$, and by the 2×2 case, they have equal determinants. Thus $\det(A^T) = \det(A)$.

General Proof (by Induction):
Base case: For $n = 1$, $A = [a]$ and $A^T = [a]$, so $\det(A) = \det(A^T) = a$. For $n = 2$, verified above.
Inductive step: Assume $\det(B^T) = \det(B)$ for all $(n-1) \times (n-1)$ matrices B.
For an $n \times n$ matrix A, expand $\det(A)$ along row i:

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} M_{ij}$$

where M_{ij} is the $(n-1) \times (n-1)$ minor obtained by deleting row i and column j.
For A^T, the element in position (j, i) is a_{ij}. Expanding $\det(A^T)$ along column i:

$$\det(A^T) = \sum_{j=1}^{n} a_{ij} C_{ji}^T = \sum_{j=1}^{n} (-1)^{j+i} a_{ij} M_{ji}^T$$

The minor M_{ji}^T (obtained by deleting row j and column i from A^T) is the transpose of M_{ij}. By the inductive hypothesis, $\det(M_{ji}^T) = \det(M_{ij})$.

Since $(-1)^{i+j} = (-1)^{j+i}$, we have:

$$\det(A^T) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(M_{ij}) = \det(A)$$

Therefore, by induction, $\det(A^T) = \det(A)$ for all $n \times n$ matrices. □

B.2 Determinant of Scalar Multiple

Theorem

For any $n \times n$ matrix A and scalar λ, $\det(\lambda A) = \lambda^n \det(A)$.

Proof:

When we multiply a matrix A by a scalar λ, each entry becomes λa_{ij}. Using the multilinearity property of determinants, we can factor out λ from each of the n rows (or columns):

$$\det(\lambda A) = \det \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{n1} & \lambda a_{n2} & \cdots & \lambda a_{nn} \end{bmatrix}$$

Factoring λ from the first row:

$$= \lambda \det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{n1} & \lambda a_{n2} & \cdots & \lambda a_{nn} \end{bmatrix}$$

Repeating this process for all n rows:

$$= \lambda^n \det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \lambda^n \det(A)$$

Therefore, $\det(\lambda A) = \lambda^n \det(A)$. □

References

[1] Alexander Panchenko. Math 448 lecture notes. Washington State University, Fall 2025. Pages 93-130.